

Full Test — Answers

Final answers with a one-line reason. For full methods, revisit the concept notes.

Objective

Q1 (b) $ab + ac$.

Q2 (b) $a^2 + 2ab + b^2$.

Q3 (b) $a^2 - b^2$.

Q4 (a) $15p^2 - 6pq$.

Q5 9801. $(100 - 1)^2$.

Q6 False. Unlike terms.

Q7 (b) $25m^2 + 60mn + 36n^2$.

Q8 (a) Both true, R explains A. $400^2 - 3^2$.

Short answer

Q9 $uv - 3u + 3v - 9$.

Q10 $36x^2 + 60x + 25$.

Q11 3025. $(50 + 5)^2 = 2500 + 500 + 25$.

Q12 Correct: $5x + 10$. Constants $-2 + 12 = 10$.

Long answer

Q13 $100ac + 10ad + 10bc + bd$.

Q14 (i) 2116 (ii) 159991. $46^2 = (40+6)^2$; $397 \times 403 = 400^2 - 3^2$.

Q15 $4y^2 + 44yz - 5y + 77z - 21$.

Q16 16×24 is larger (384 vs 364). $(20-4)(20+4) = 384$; $(20-6)(20+6) = 364$.

Total: 28 marks.